

Semantic Search: An Overview

Traditional keyword search matches the exact words of a query against an inverted index. It is fast and predictable, but it fails when the user and the author describe the same idea with different words. A query for "car insurance" will miss a document that only talks about "vehicle coverage". Semantic search addresses this by comparing meaning instead of vocabulary.

The core idea is to represent every piece of text as a dense vector, called an embedding. An embedding model is trained so that passages with similar meaning are mapped to nearby points in a high-dimensional space, while unrelated passages end up far apart. Once documents and queries live in the same space, retrieval becomes a geometric problem: find the stored vectors closest to the query vector.

How Embeddings Are Produced

Modern embedding models are transformer encoders. The input text is split into sub-word tokens, each token is turned into a contextual representation by several layers of self-attention, and the token representations are pooled, usually by averaging, into a single fixed-size vector. Small models such as all-MiniLM-L6-v2 produce 384-dimensional vectors and run comfortably on a CPU.

Encoders have a maximum input length, measured in tokens rather than words. Text beyond that limit is silently truncated, which is why long documents are split into chunks before they are embedded. Good chunks respect sentence boundaries and overlap slightly, so that an idea spanning two chunks is still fully present in at least one of them.

Similarity and Vector Databases

The most common similarity measure is cosine similarity, which compares the angle between two vectors and ignores their length. When vectors are normalised to unit length, cosine similarity is simply their dot product. Scores near one indicate closely related text, while scores near zero indicate unrelated text.

Comparing a query against millions of vectors one by one would be slow, so vector databases build approximate nearest-neighbour indexes. HNSW, a graph of vectors linked to their neighbours, lets a search hop towards the most similar items in logarithmic time. Vector databases also store a payload next to each vector, such as the source document, page number and original text, so that results can be shown to the user and filtered.